

SUMMARY TABLE: BIOLOGY GRADE 11

Sub-Strand 2.3: Excretion in Plants — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.3: Excretion in Plants
Driving Question	DRIVING QUESTION: How do trees and crops clear out their metabolic wastes without any organs or excretory systems, and why are these waste products worth so much money in our markets?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Where Do Plant Wastes Go? Gum, Droplets and Falling Leaves	We observed gum or resin and milky latex oozing from cut Acacia and Euphorbia, droplets along the edges of grass and arrowroot leaves in the early morning, and yellow leaves falling from trees. The chart showed plants have much lower metabolic rates than animals, and the mesophyll diagrams showed CO ₂ and O ₂ being passed between respiration and photosynthesis.	Excretion is the removal of metabolic wastes from the body. Plants lack complex excretory organs like kidneys because (1) their metabolic rate is low so they make little waste, (2) they recycle CO ₂ and O ₂ between respiration and photosynthesis, and (3) they store insoluble wastes such as resin, latex and gum in tissues like bark, vacuoles and old leaves.	Claim: Plants do not need kidneys. Evidence: low metabolic rates, recycling of CO ₂ and O ₂ in mesophyll cells, and stored products like gum arabic and latex that ooze out when tissue is cut. Reasoning: little waste is made and much of it is reused or stored harmlessly, so no complex organ is needed. Still unexplained: how the remaining wastes actually leave the plant, such as the leaf-edge droplets, and why some products are valuable in Kenyan markets.
Lesson 2 How Do Plants Get Rid of Wastes? Nature Walk, Exudation and Guttation Evidence	We observed hardened amber gum on Acacia and milky latex oozing from a cut Euphorbia (exudation); lenticels on stems; yellow fallen leaves and leaf scars (tissue fall); droplets at the edges of grass and arrowroot leaves, and droplets forming at leaf margins of the potted seedling under the bell jar compared with the control (guttation); water vapour droplets in the plastic bag over a branch (transpiration).	Plants eliminate gaseous and liquid wastes by diffusion through stomata and lenticels, transpiration (water vapour), exudation (gums, resins, latex), guttation (liquid droplets from hydathodes in humid conditions), and by shedding leaves, bark and fruits. Some of these products, such as gum arabic and latex, have commercial value in Kenyan markets.	Plants have no kidneys, but their wastes leave through several surfaces and structures: gases and water vapour diffuse out through stomata and lenticels, while liquid wastes and dissolved substances leave by guttation or exudation. This explains the oozing sap and leaf-edge droplets seen locally. However, we cannot yet explain what happens to insoluble wastes that cannot evaporate or ooze out; this is the question for Lesson 3.
Lesson 3	We collected young green and old	Plants deposit insoluble metabolic wastes	Claim: Plants excrete insoluble wastes by

Where Do the Wastes Go? Shedding Old Leaves and Locking Away Crystals	yellow/brown leaves from school trees and compared them with hand lenses. Under the light microscope we found [describe crystal shape, how many in young versus old leaves], and recorded how crystals reacted to iodine and dilute acid. We also noted resin or gum on the bark of Acacia and Euphorbia and set up the guttation experiment.	such as calcium oxalate crystals, tannins and resins in non-vital organs (bark, old leaves, fruits and flowers). These parts are shed periodically through tissue fall. Plants therefore do not need kidneys because wastes are stored in harmless forms.	depositing them in old or non-vital tissues that are then shed. Evidence: old yellow leaves had more crystals than young leaves, and the crystals did not dissolve in ethanoic acid (or reacted in the specific way recorded). Reasoning: locking wastes away in tissue the plant no longer needs, then dropping it, removes the wastes without an excretory system, which also explains the gum and resin on cut Acacia and Euphorbia and the falling leaves.
Lesson 4 Cash from Waste: Categorising Plant Excretory Products and Their Value in Kenya	We observed hardened gum and resin on Acacia bark, Euphorbia latex (teacher-handled), and the smell of essential oils in crushed lemongrass or eucalyptus leaves, and we researched products such as rubber, quinine, papain, tannins, caffeine and gum arabic.	We learned that plant excretory products fall into types: latex, tannins, alkaloids, gums and resins, and essential oils. Each comes from particular plants and has uses in medicine, industry and food, and many are sold in Kenyan markets, for example tea from Kericho, gum arabic from northern Kenya, and quinine for malaria.	Plants lack kidneys, but they store or release waste chemicals in latex vessels, bark, leaves and sap. Humans use the chemical properties of these substances, so what is waste to the plant is valuable to people, which answers why Acacia and Euphorbia products are worth money and leads us to ask how communities use them and whether any are harmful.
Lesson 5 Who Profits from Plant Wastes? Community Uses, Alkaloid Risks and the Final Explanation	What did we observe and record? (e.g. oozing gum from Acacia and latex from Euphorbia, guttation droplets on arrowroot and grass leaves, fallen old leaves, and the uses, prices and risks of plant products reported by community members in our interviews and portfolios)	What did we learn? (e.g. plants lack excretory organs because they produce few toxic wastes, recycle gases, and store wastes in dead tissues; wastes leave by diffusion, transpiration, guttation, exudation and organ fall; products such as gum arabic, latex, resins, tannins, quinine, caffeine and essential oils are used for medicine, sealing, tanning and drinks and bring income; some alkaloids are harmful when abused)	How can we explain the phenomenon in our own words? (Write a Claim-Evidence-Reasoning explanation of how trees and crops clear their wastes without kidneys and why these wastes are valuable in Kenyan markets, and state the new question: How can sustainable harvesting of useful plant excretory products be balanced with environmental conservation?)

ATTRIBUTION AND LICENSING

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